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STM32 NUCLEO

Périphérique GPIO

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STM32 Nucleo Board

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Documentations techniques utiles

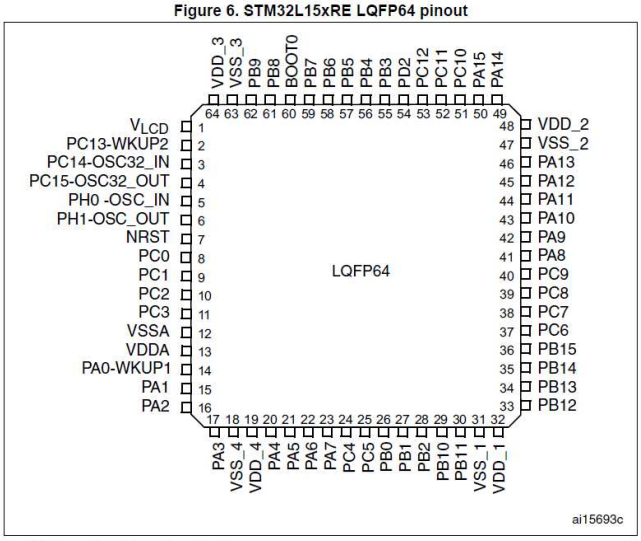
PM0056 Programming manual
STM32F10xxx/20xxx/21xxx/L1xxxxCortex-M3 programming manual

RM0038 Reference manual
STM32L100xx, STM32L151xx, STM32L152xx and STM32L162xx advanced ARM-based 32-bit MCUs

STM32L152RE

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Figure 6. STM32L15xRE LQFP64 pinout



1. This figure shows the package top view.

Présentation du STM32L152RE

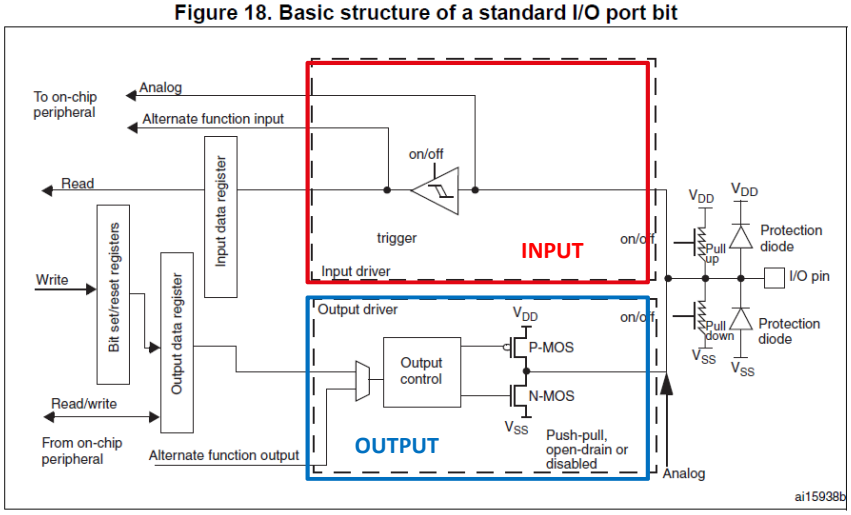
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- Le STM32L152RE dispose de 51 GPIOs répartis selon 4 ports:
  - 3 ports A, B, C comportant chacun 16 GPIOs noté PAx ou PBx ou PCx
  - 1 port H composé de 2 GPIOs PH0 et PH1
  - 1 port D composé d'un seul GPIO PD2
- Chaque GPIO peut être configuré par le software en entrée , en sortie, en analogique ou en fonction multiplexée grâce à un registre de configuration (32 bits).
- Les GPIO peuvent être partagés avec des fonctions multiplexées c'est-à-dire utilisés par d'autres périphériques que le port.

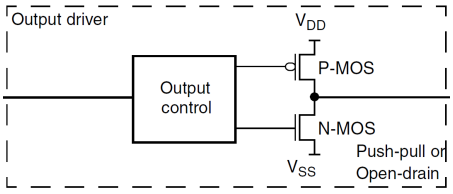
Structure d'un port GPIO

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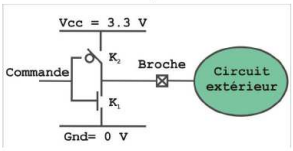
Figure 18. Basic structure of a standard I/O port bit



Output Mode – Open Drain et Push-pull

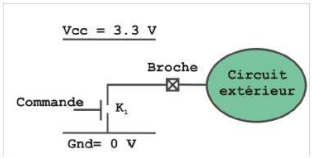


PUSH PULL



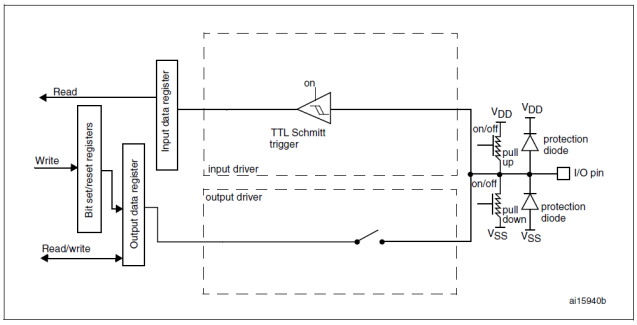
Les transistors N et P sont activés. Le potentiel de la broche vaut Vcc ou GND.
Le GPIO impose le potentiel.

OPEN-DRAIN

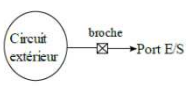


Seul le transistor N est activé. Le port ne peut que forcer le potentiel de la broche à GND.
Si l'interrupteur est ouvert, le circuit extérieur fixe le potentiel de la broche.

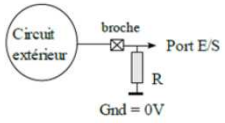
Input Mode – Floating input et Pull Down/Up input



Floating Input

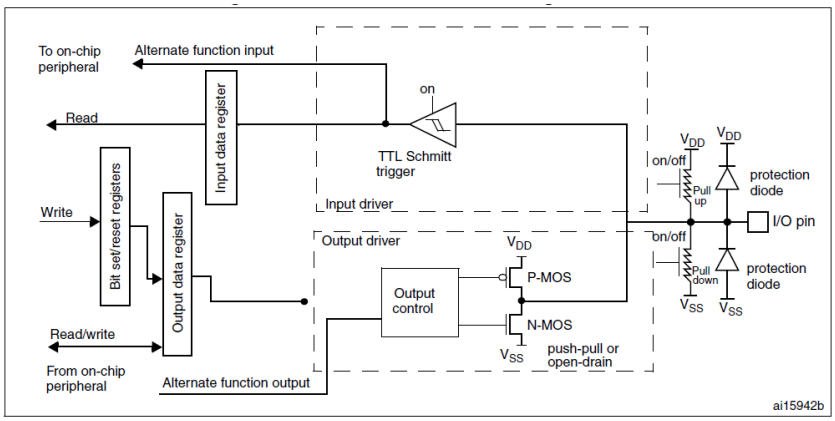


Pull Down/ Pull Up Input



Fonction Multiplexée

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Fonction Multiplexée

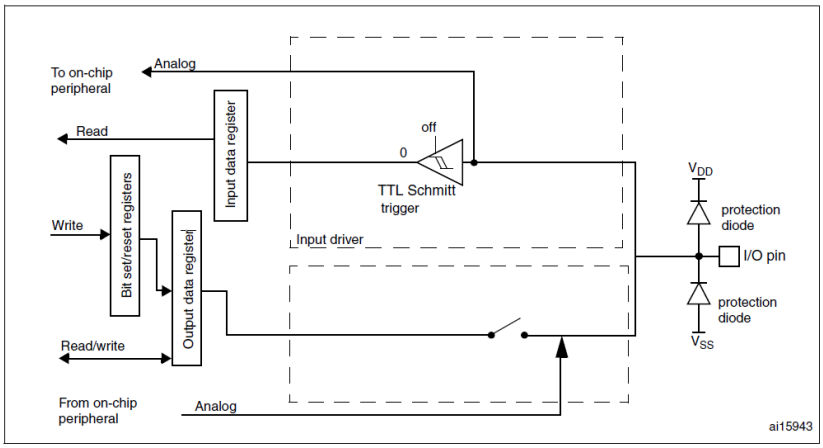
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Table 8. STM32L151xE and STM32L152xE pin definitions

Pins					Pin name	Pin Type(1)	I / O structure	Main function(2) (after reset)	Alternate functions
LQFP144	UFBGA132	LQFP100	LQFP64	WLCSP104					
6	E2	6	1	E7	V_{LCD}(3)	S	-	V_{LCD}	-
7	C1	7	2	C9	PC13-WKUP2	I/O	FT	PC13	WKUP2/RTC_TAMP1/RTC_TS/RTC_OUT
8	D1	8	3	D8	PC14-OSC32_IN(4)	I/O	TC	PC14	OSC32_IN

Entrée/Sortie Analogique

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Configuration d'un port GPIO

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A Chaque port sont associés différents registres permettant de configurer chaque GPIO de manière indépendante. On trouve:

- Registres de configuration :
  - Mode: Ouput, Input, Analog, Alternative Functions
  - Options supplémentaires: Pull up/down, floating, push pull, open drain, speed
- Registres de données

Table des configurations possibles d'un GPIO

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Nom des registres

Table 36. Port bit configuration table[1]

MODER(i) [1:0]	OTYPER(i)	OSPEEDR(i) [B:A]	PUPDR(i) [1:0]	I/O configuration	
01	0	SPEED [B:A]	0 0	GP output	PP
	0		0 1	GP output	PP + PU
	0		1 0	GP output	PP + PD
	0		1 1	Reserved	
	1		0 0	GP output	OD
	1		0 1	GP output	OD + PU
	1		1 0	GP output	OD + PD
	1		1 1	Reserved (GP output OD)	
10	0	SPEED [B:A]	0 0	AF	PP
	0		0 1	AF	PP + PU
	0		1 0	AF	PP + PD
	0		1 1	Reserved	
	1		0 0	AF	OD
	1		0 1	AF	OD + PU
	1		1 0	AF	OD + PD
	1		1 1	Reserved	
00	x	x x	0 0	Input	Floating
	x	x x	0 1	Input	PU
	x	x x	1 0	Input	PD
	x	x x	1 1	Reserved (input floating)	
11	x	x x	0 0	Input/output	Analog
	x	x x	0 1	Reserved	
	x	x x	1 0		
	x	x x	1 1		

1. GP = general-purpose, PP = push-pull, PU = pull-up, PD = pull-down, OD = open-drain, AF = alternate function.

OUTPUT

Alternative Function

INPUT

Analog

Registre de configuration du MODE: GPIOx_MODER

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7.4.1 GPIO port mode register (GPIOx_MODER) (x = A..H)

Address offset: 0x00

Reset values:

- 0xA800 0000 for port A
- 0x0000 0280 for port B
- 0x0000 0000 for other ports

Px9

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Px6 Px4 Px2 Px0

Bits 2y:2y+1 **MODERy[1:0]:** Port x configuration bits (y = 0..15)
These bits are written by software to configure the I/O direction mode.
00: Input (reset state)
01: General purpose output mode
10: Alternate function mode
11: Analog mode

Registre de configuration du type de sortie: GPIOx_OTYPER

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7.4.2 GPIO port output type register (GPIOx_OTYPER) (x = A..H)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OT15	OT14	OT13	OT12	OT11	OT10	OT9	OT8	OT7	OT6	OT5	OT4	OT3	OT2	OT1	OT0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **OTy**: Port x configuration bits (y = 0..15)
These bits are written by software to configure the output type of the I/O port.
0: Output push-pull (reset state)
1: Output open-drain

Registre de configuration de la vitesse: GPIOx_OSPEEDR

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7.4.3 GPIO port output speed register (GPIOx_OSPEEDR) (x = A..H)

Address offset: 0x08

Reset values:

- 0x0000 00C0 for port B
- 0x0000 0000 for other ports

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OSPEEDR15 [1:0]		OSPEEDR14 [1:0]		OSPEEDR13 [1:0]		OSPEEDR12 [1:0]		OSPEEDR11 [1:0]		OSPEEDR10 [1:0]		OSPEEDR9 [1:0]		OSPEEDR8 [1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OSPEEDR7[1:0]		OSPEEDR6[1:0]		OSPEEDR5[1:0]		OSPEEDR4[1:0]		OSPEEDR3[1:0]		OSPEEDR2[1:0]		OSPEEDR1 [1:0]		OSPEEDR0 1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 2y:2y+1 **OSPEEDRy[1:0]:** Port x configuration bits (y = 0..15)
These bits are written by software to configure the I/O output speed.
00: Very low speed
01: Low speed
10: Medium speed
11: High speed

Note: Refer to the product datasheets for the values of OSPEEDRy bits versus V_{DD} range and external load.

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Registre de configuration de la résistance de rappel: GPIOx_PUPDR

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7.4.4 GPIO port pull-up/pull-down register (GPIOx_PUPDR) (x = A..H)

Address offset: 0x0C

Reset values:

- 0x6400 0000 for port A
- 0x0000 0100 for port B
- 0x0000 0000 for other ports

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PUPDR15[1:0]		PUPDR14[1:0]		PUPDR13[1:0]		PUPDR12[1:0]		PUPDR11[1:0]		PUPDR10[1:0]		PUPDR9[1:0]		PUPDR8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PUPDR7[1:0]		PUPDR6[1:0]		PUPDR5[1:0]		PUPDR4[1:0]		PUPDR3[1:0]		PUPDR2[1:0]		PUPDR1[1:0]		PUPDR0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 2y:2y+1 **PUPDRy[1:0]:** Port x configuration bits (y = 0..15)
These bits are written by software to configure the I/O pull-up or pull-down
00: No pull-up, pull-down
01: Pull-up
10: Pull-down
11: Reserved

Registre de donnée du mode INPUT: GPIOx_IDR

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7.4.5 GPIO port input data register (GPIOx_IDR) (x = A..H)

Address offset: 0x10

Reset value: 0x0000 XXXX (where X means undefined)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDR15	IDR14	IDR13	IDR12	IDR11	IDR10	IDR9	IDR8	IDR7	IDR6	IDR5	IDR4	IDR3	IDR2	IDR1	IDR0
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **IDRy**: Port input data (y = 0..15)
These bits are read-only and can be accessed in word mode only. They contain the input value of the corresponding I/O port.

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Registre de donnée du mode OUTPUT:
GPIOx_ODR

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7.4.6 GPIO port output data register (GPIOx_ODR) (x = A..H)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODR15	ODR14	ODR13	ODR12	ODR11	ODR10	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **ODRy**: Port output data (y = 0..15)

These bits can be read and written by software.

Note: For atomic bit set/reset, the ODR bits can be individually set and reset by writing to the GPIOx_BSRR register (x = A..H).

Registre de mise à 1 ou à 0 en mode sortie:
GPIOx_BSRR

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7.4.7 GPIO port bit set/reset register (GPIOx_BSRR) (x = A..H)

Address offset: 0x18

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:16 **BRy:** Port x reset bit y (y = 0..15)

These bits are write-only and can be accessed in word, half-word or byte mode. A read to these bits returns the value 0x0000.

0: No action on the corresponding ODRx bit

1: Resets the corresponding ODRx bit

Note: If both BSx and BRx are set, BSx has priority.

Bits 15:0 **BSy:** Port x set bit y (y= 0..15)

These bits are write-only and can be accessed in word, half-word or byte mode. A read to these bits returns the value 0x0000.

0: No action on the corresponding ODRx bit

1: Sets the corresponding ODRx bit

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Registre de sélection de la fonction multiplexée: GPIOx_AFRL

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7.4.9 GPIO alternate function low register (GPIOx_AFRL) (x = A..H)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFRL7[3:0]				AFRL6[3:0]				AFRL5[3:0]				AFRL4[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFRL3[3:0]				AFRL2[3:0]				AFRL1[3:0]				AFRL0[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **AFRLy:** Alternate function selection for port x bit y (y = 0..7)
These bits are written by software to configure alternate function I/Os

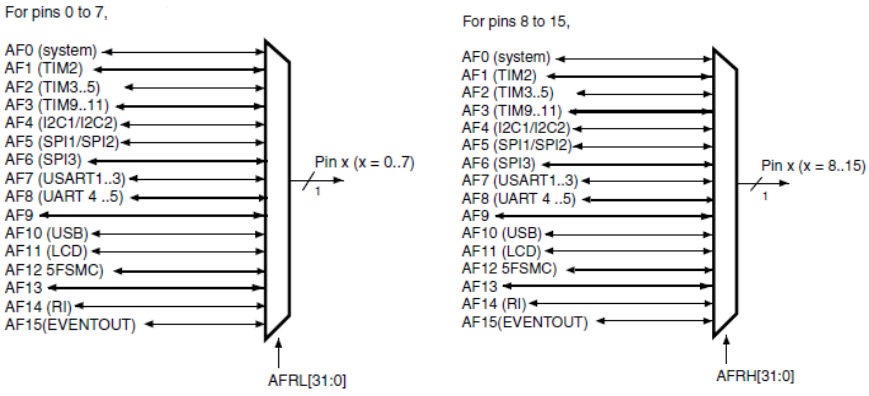
AFRLy selection:

0000: AF0	1000: AF8
0001: AF1	1001: AF9
0010: AF2	1010: AF10
0011: AF3	1011: AF11
0100: AF4	1100: AF12
0101: AF5	1101: AF13
0110: AF6	1110: AF14
0111: AF7	1111: AF15

Il existe aussi GPIOx_AFRH pour les GPIO N° 8 à 15.

Registre de sélection de la fonction multiplexée: GPIOx_AFRL (suite)

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